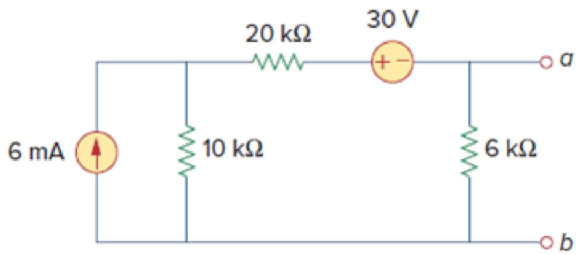
Chapter 4, Problem 37P

Problem

Find the Norton equivalent with respect to terminals *a-b* in the circuit shown in Fig. 4.104.

Figure. 4.104:



Step-by-step solution

Step 1 of 6

Refer to circuit diagram in Figure 4.104 in the text book.

To calculate the short circuit current, short circuit the terminals *a* and *b*. The modified circuit is shown in Figure 1.

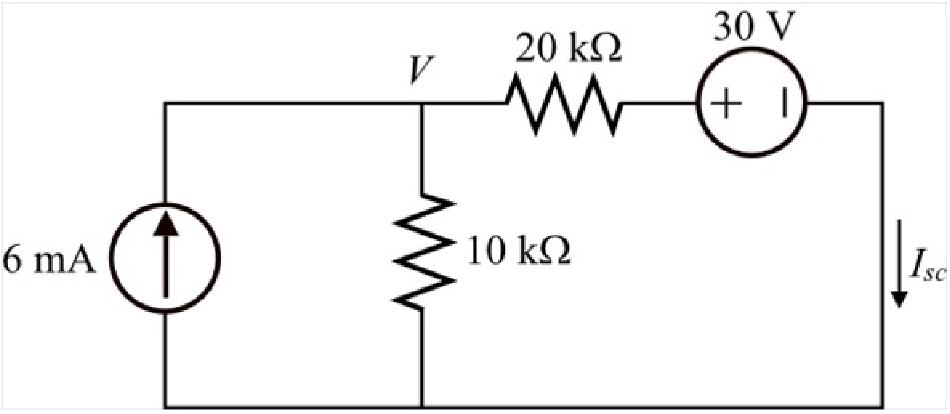


Figure 1

Step 2 of 6

Apply Kirchhoff's Current Law at node 1.

$$\frac{V}{10\text{ k}}+\frac{V-30}{20\text{ k}}-6\text{ m}=0$$

$$\frac{2V+V-30-120}{20\text{ k}}=0$$

$$2V+V-30-120=0$$

$$3V-150=0$$

$$3V=150$$

$$V=\frac{150}{3}$$

$$V=50\text{ V}$$

Step 3 of 6

Calculate the value of the Norton's current I_N.

$$I_N = \frac{V - 30}{20\text{ k}}$$

Substitute 50 V for V.

$$\begin{aligned} I_N &= \frac{50 - 30}{20\text{ k}} \\ &= \frac{20}{20\text{ k}} \\ &= 1\text{ mA} \end{aligned}$$

Therefore, the value of the Norton's current I_N is $\boxed{1\text{ mA}}$.

Step 4 of 6

To calculate the Thevenin's resistance, open circuit the current source and short circuit the voltage source. The modified circuit is shown in Figure 2.

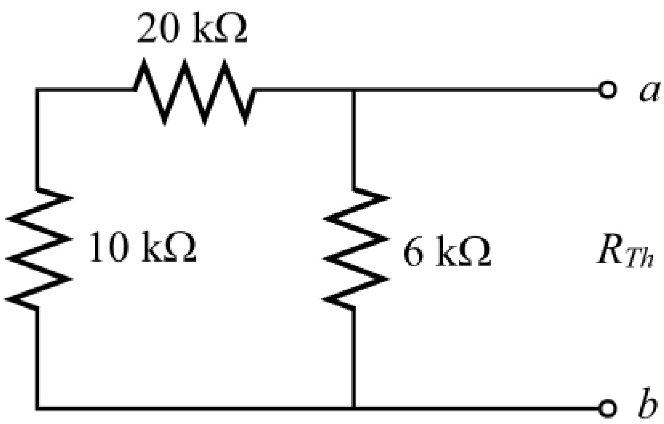


Figure 2

[Comments (2)]

Anonymous

where did the 6k come from?

Anonymous

its in the problem...

Step 5 of 6

Calculate the value of the Norton's resistance R_N.

$$\begin{aligned} R_N &= (10\text{ k}\Omega + 20\text{ k}\Omega) \,||\, (6\text{ k}\Omega) \\ &= (30\text{ k}\Omega) \,||\, (6\text{ k}\Omega) \\ &= \frac{(30\text{ k}\Omega)(6\text{ k}\Omega)}{30\text{ k}\Omega + 6\text{ k}\Omega} \\ &= \frac{180\text{ k}\Omega}{36} \end{aligned}$$

$$= 5\text{ k}\Omega$$

Therefore, the value of the Norton's resistance R_N is $\boxed{5\text{ k}\Omega}$.

Step 6 of 6

Draw the Norton’s equivalent circuit diagram as shown in Figure 3.

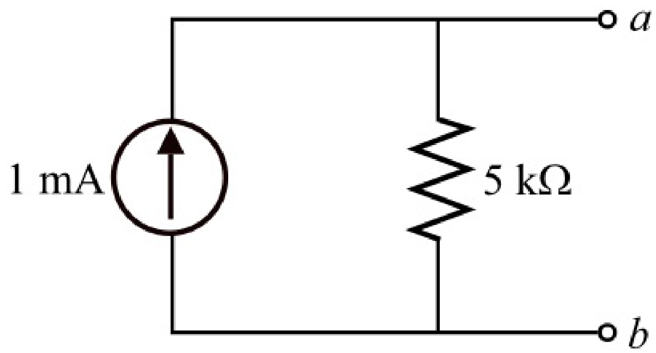


Figure 3

Therefore, the Norton’s equivalent circuit is shown in Figure 3.